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Coursework 2 (60%) Applied econometrics EC3017
(Words count: 1769 words)

Introduction (632 words)

This report evaluates the impact of a campaign promoting family-friendly workplaces on better labour market outcomes for women. To understand the importance of this topic, we need to examine women's integration into the US labour market. According to the US Bureau of Labour Statistics, women's labour force participation sharply increased throughout the 1960s, reaching a peak of 60.0% in 1999. Today, women's labour market participation represents almost half, but the quest for equal representation is now stagnating. Adapting the work environment is a major concern. The other key interest of this report is the impact of information campaigns. These campaigns are one of the strategies governments use to influence people's behaviour towards work. These campaigns are important to assess because they are costly, and assessing their effectiveness is important.

This report will attempt to determine, using the Difference-in-differences method, whether we can demonstrate the campaign's usefulness and measure its impact.

This report also analyses the impact of a potential increase in the labour force, including its effect on the wages and total income of women in the US, and whether this evolution follows the standard pattern of the rest of the non-impacted areas, or differs. This analysis will also give us the opportunity to evaluate the resemblance of our results to the theory of Labour supply and Demand theory.

The hypothesis we will assess is that an information campaign can have a real impact on women's employment rates and on the probability of being employed in the US. The expected reaction to this campaign would be a positive labour supply shock. The reasoning is that this theory is based on the potential increase in a family-friendly work environment; this increase would lead to an increase in utility provided at work because of the presence of a non-pecuniary amenity.

According to labour economics literature, the choice a worker has to work or not is based on the utility he gets from working, if we perceive the existence of a family-friendly work environment as a positive amenity, then the women who will get to decide if she enters the workforce or not will derive their utility non only from the wage but also from this amenity. The model we are describing here could be illustrated like this:

$$U = f(w, a)$$

Where w describes the wage and a the amenities the worker gets.

Such a campaign could increase the perceived value of those amenities and make companies that propose it more attractive to women. It could also increase the number of women in the total workforce by the utility a female worker gets from working; this potential supply shock would result in a shift to the right of the supply curve. This shift to the right could also come from an increase in the number of hours women work.

Consequently, we can draw two hypotheses:

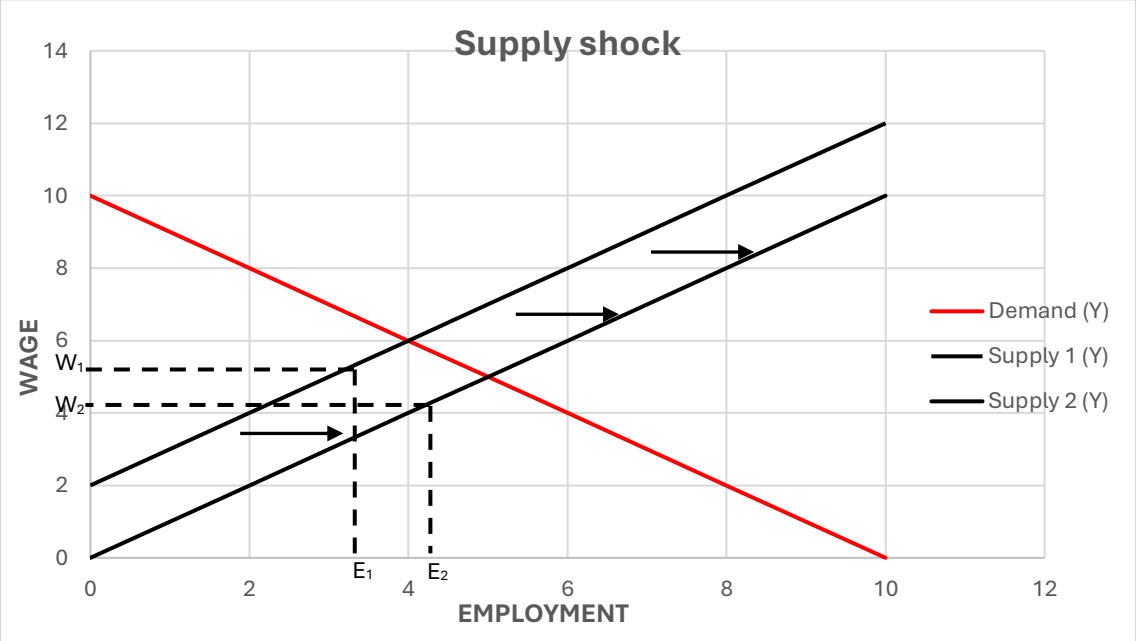
Hypothesis 1: Employment effect, the equilibrium of the labour market shifts, increasing the employment rate and the number of working hours.

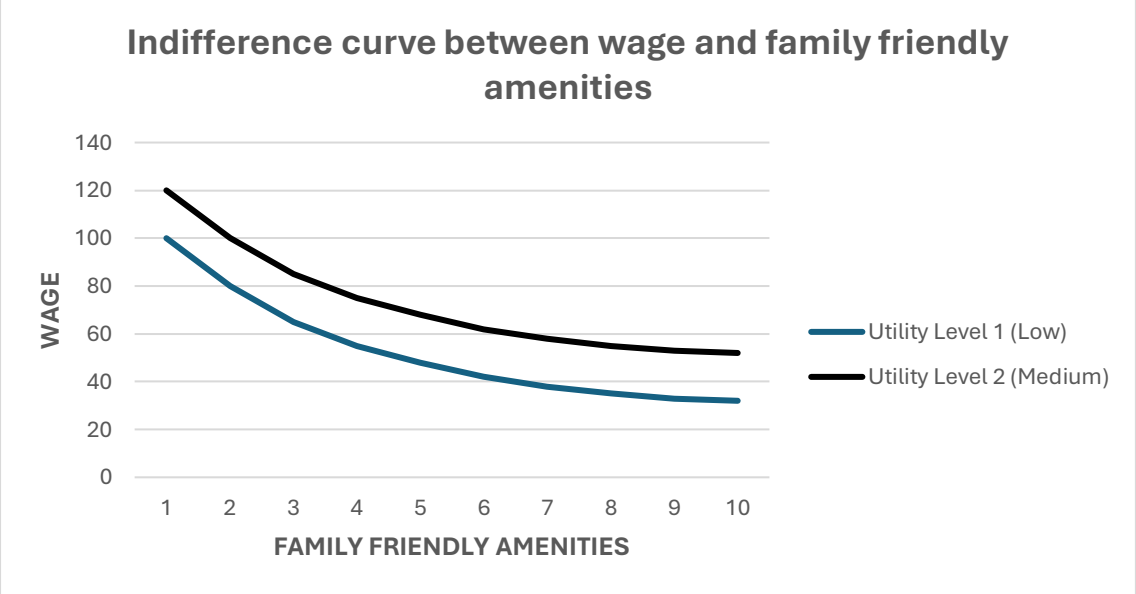
Hypothesis 2: Wage effect, assuming that demand remains relatively inelastic in the short run, is that an increase in labour supply would have a negative impact on the wage equilibrium, potentially lowering the hourly rate.

Hypothesis 3 (Bonus): The presence of amenities positively affects the utility workers' utility from their job. The expected effect would be that companies would trade some of their salary against amenities. This effect can be observed in the health care benefits employers provide to their employees. According to Labour Economics (George J. Borjas), the average wage of someone who receives such amenities is lower than that of someone who receives no amenities.

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This analysis of the labour market is based on the compensating wage differential introduced by Adam Smith in his book The Wealth of Nations. Here are some graphical representations of our hypothesis:





Data Section: (147 words)

Our primary analysis is based on the "CW2_main_2025_26_core.dta" dataset, which contains 191,020 observations of Women in the United States. The data cover the period between 2005 and 2014. This allows us to assess the impact of the 2009 information campaign. To assess the parallel trends, we will also use an extended version of this database.

Variable	Label	Mean	Std.Dev	Min	Max
Employment	1 if employed	0.82	0.38	0	1
Annual Hours	Total hours worked last year	1,455.11	1,057.51	0	5,298
Hourly Wage	Hourly wage rate in dollar	14.85	8.95	0.66	65.12

Labour Income	Total annual labor income in dollar	32,859.03	20,909.18	1,283.75	222,3
Key predictors					
Family friendly	1 if the workspace is FF	0.48	0.50	0	1
Post-policy	1 if year superior or equal to 2010	0.47	0.50	0	1
Controls					
Age	Potential experience in Years	16.79	9.72	-5	47
Married	1 if married	0.73	0.44	0	1
Children < 5	Number of children under 5	0.35	0.64	0	7
Children 5-17	Number of children aged 5-17	0.90	1.08	0	9
Education	Categorical (1=HS...5=Adv)	3.04	1.17	1	5

The main dependent variable of interest is employment status, a dummy variable indicating whether the worker is employed. We are also going to use total hours worked, conditional on employment, and earnings responses, using Hourly Wages and Total Labour Income.

The main independent variable is a dummy variable indicating whether the individual works in a family-friendly environment. We will control for demographic and human capital variables, such as education, age (as a proxy for potential experience), we also use age squared as a proxy of experience squared, race (to account for potential discrimination). Marital status (to be able to account for the dual income), and number of children, because they are important determinants of female labour supply.

Methodology Section : (317 words)

To evaluate the 2009/2010 information campaign on the women's labour market, we use Difference-in-Differences (DiD). We use this method to isolate the effect of the information campaign on women's labour force participation by comparing the evolution of the treated and non-treated groups.

Regression 1

The first part of our work was to estimate the baseline regression:

$$y_{it} = \alpha FF_i + \beta(FF_i * Post_t) + \tau + \theta_s + \epsilon_{it}$$

Regression 2

To obtain a more precise estimate, we included a vector of individual-level controls, such as age, education, marital status, and the number of children:

$$y_{it} = \alpha FF_i + \beta(FF_i * Post_t) + \beta X'_{ist} + \tau + \theta_s + \epsilon_{it}$$

Where $\beta X'_{ist}$ is the vector of controlled variables **(nchlt5), (nchlt517), (married),** race ethnicity, **(educ)**, **(pexp)** and **(pexp2).** And we estimate this regression for four outcome variables: employment, Annual Hours Worked, Hourly Wages, and Total Labour Income.

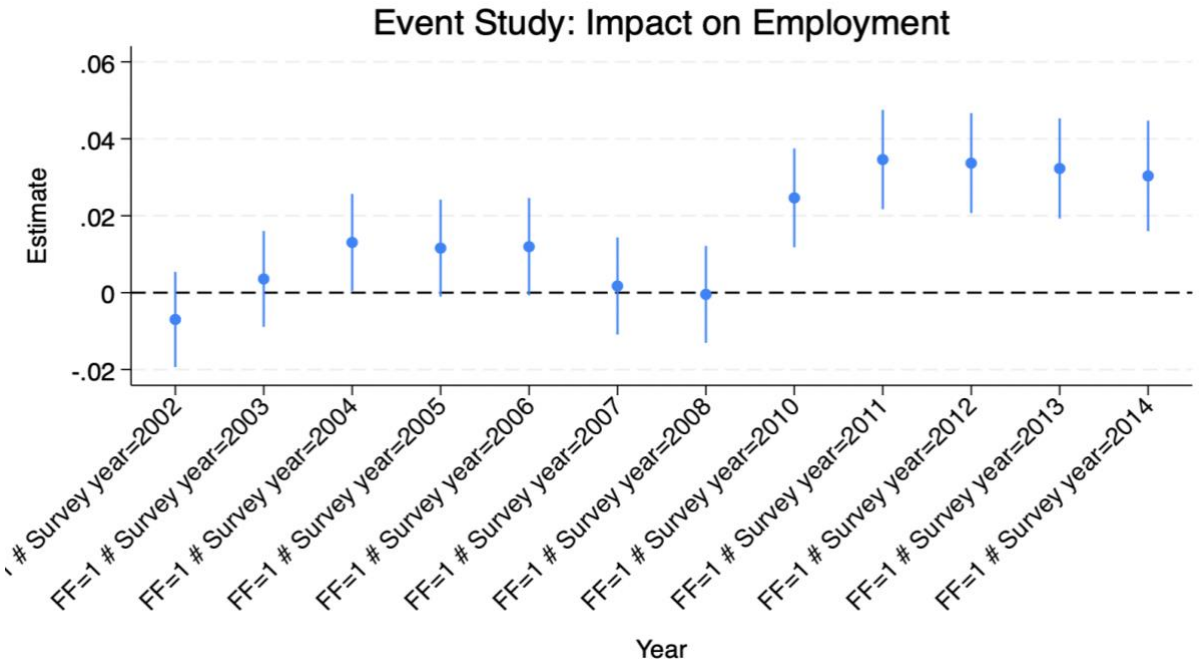
This methodology is best suited because it enables us to isolate the campaign's effect by accounting for human capital variables. The DiD methodology also allows us to test the validity of parallel trends before the policy's implementation, ensuring that both samples were moving in the same direction. To account for heteroskedasticity, we used a robust standard error.

The causal validity of this DiD relies on the Parallel trends assumption, which holds that, in the absence of intervention, both groups' averages would have followed the same trend over time.

Since we cannot observe what did not happen, we assess the probability that they follow the same trend by observing their trends before treatment. That's why we use the extended dataset. We interact the treatment indicator (FF) with a full set of year dummies, omitting 2009 as the reference base year:

$$yit = \sum_{j \neq 2009} nj\left(FF_i \times Year_j\right) + \alpha_i + \tau_t + \epsilon_{it}$$

The result of this test is in the figure below:



The pre-treatment coefficients are statistically insignificant, suggesting no differential trends prior to treatment in the year (eg., 2007 and 2008) with p-values of 0.787 and 0.945, respectively. This provides supportive evidence for the parallel trends assumption.

Result Section :

Dependent variable:	1	2	3	4	5
	Employment	Employment	Annual hours	Hourly wage	Labour Income
	Baseline Regression	**Regression 2**	**Regression 2**	**Regression 2**	**Regression 2**
DiD Estimator	0.0261	0.0270	110.15	-0.736	-138.88
(FF X Post)	(0.0030)	(0.0029)	(8.942)	(0.0864)	(202.02)
Percentage effect		+3.3%	+7.4%	-5.2%	-0.4%

Observations	191,02	191,02	191,02	124,527	124,527
R-squared	0.210	0.262	0.152	0.284	0.287
Controls	No	Yes	Yes	Yes	Yes

Interpretation of the results: (305 words)

In Column (2), we can observe the impact of the campaign on employment. The coefficient (0.0270) indicates that the campaign had a positive effect on employment for women in a family-friendly environment relative to the control group ($p<0.01$, so the result is statistically significant. This result controls for individual-level variables that constitute human capital. We can also say that even without controlling for those variables, the results in Column (1) are similar. This result aligns with our expectations. Now we can see whether the increase in the workforce created the effect we expected.

In Column (3), we observe the campaign's impact on annual hours worked. The coefficient (110.15) indicates that the campaign increases labour force intensity by increasing the number of hours worked per year. Compared with the pre-treatment period, labour supply increases by 7.4%. This increase in hours worked could be related to the rise in the utility that workers derive from the growing importance of “family-friendly” amenities.

In Column (4), we observe a statistically significant negative effect on the hourly wage. This equates to a 5.2% decrease relative to the baseline. Although this effect may be considered small, it aligns with the economic theory of wage evolution, as a shift in the supply of labour reduces the wage equilibrium.

Although we still see in Column (5) that total labour income slightly decreased, the coefficient is non-significant because it was completely offset by the increase in the number of hours worked, which significantly increased.

Overall, those results align with our hypothesis. The consequence of the campaign is that family-friendly workplaces are more attractive (increasing non-pecuniary utility). The policy shifted the labour supply curve to the right, and because labour demand is relatively inelastic in the short run, the supply extension shifted the equilibrium to a lower wage.

Conclusion: (368 words)

The conclusion of this research shows that the information campaign had some effect on female workers. The effect of this campaign aligns with the economic theory stated in the introduction. Regarding the implications of female participation in the workforce, we observe an increase that aligns perfectly with the utility of workers and the non-pecuniary nature of a family-friendly workspace, and this is reinforced by the increase in hours worked. Regarding the effect on female workers' wages, the hourly wage coefficient results support the second and third hypotheses. The indication that labour income slightly decreased is non-significant because workers work more, so the effect on total income is offset. Those findings enable us to take seriously the three hypotheses that we presented in the introduction. The role of amenities is clear in the reaction to the policy, and the market reaction in terms of wages and employment is also aligned with our first intuition.

We would need to improve our model or have longer-term data to draw firm conclusions about the evolution of wages. The fact that we are not able to see the long-term impact limits our use, because we cannot see whether wage decreases then evolve over the long term, or whether the hourly wage equilibrium increases. Another limitation of the information campaign is that it may affect how people view a family-friendly workplace and might increase their preference for it, but

it does not force or incentivise companies to change. This is one of the major limitations of an information campaign as a solution to change a market where there is no obligation to change anything. Other measures that would either force or incentivise firms to become more family-friendly could have greater impact. Another limitation of our analysis is that we did not analyse who was impacted by the information campaign, whether it reached only the upper class, and whether there were disparities in the impact of this campaign.

Overall, even if information campaigns can have a real impact on the workspace and on decisions, one could argue that they are useful in changing the way people think so that policies can be implemented that would constrain companies to adopt certain uses in their workplaces.

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Appendix 1: STATA:

```
*Do file Applied econometrics
describe
summarize
*creation of an interaction term (Difference-in-Differences estimator)
generate did_term = FF * post
*Creating a baseline regression by taking into account the fact that statefip is a
dummy variable and year is also a dummy variable.
regress myemp FF did_term i.year i.statefip, robust
*Regression with controle of other varible
regress myemp FF did_term i.year i.statefip nchlt5 nch517 married i.race_ethnicity
i.educ pexp pexp2, robust

*Step 3: Hours Worked & Percentage Calculation
*Run the regression
regress hours_last_year FF did_term i.year i.statefip nchlt5 nch517 married
i.race_ethnicity i.educ pexp pexp2, robust
* 2. Calculate the Baseline Mean (The Denominator)
* Average hours for FF workplaces (FF=1) BEFORE the policy (post=0)
summarize hours_last_year if FF == 1 & post == 0
*Regression for wage
regress wage FF did_term i.year i.statefip nchlt5 nch517 married i.race_ethnicity
i.educ pexp pexp2, robust
summarize wage if FF == 1 & post == 0
*Regresion for Income
regress income FF did_term i.year i.statefip nchlt5 nch517 married i.race_ethnicity
i.educ pexp pexp2, robust
summarize income if FF == 1 & post == 0

*Paralele trends for the extended data base
use "CW2_main_2025_26_extended.dta", clear
regress myemp i.FF##ib2009.year i.statefip, robust
* Look at the coefficients for 1.FF#2005.year, 1.FF#2006.year, etc.
* They should be statistically insignificant (close to 0).
coefplot, keep(1.FF#*) vertical yline(0) drop(_cons)

*Laste step Descriptive Statistics Table
* Use the 'summarize' command on the specific variables you used
summarize myemp hours_last_year wage income FF post nchlt5 married educ

* 1. Load the EXTENDED dataset
use "CW2_main_2025_26_extended.dta", clear

* 2. Run the Event Study Regression
* We interact FF with EVERY year, using 2009 as the reference point.
* The coefficients for years 2005, 2006, 2007, 2008 tell us if trends were parallel.
regress myemp i.FF##ib2009.year i.statefip, robust

* 3.
* If this fails, just share the regression table results.
ssc install coefplot
coefplot, keep(1.FF#*) vertical yline(0) drop(_cons) ///
  xline(2009.5) ///
  title("Event Study: Impact on Employment") ///
  xtitle("Year") ytitle("Estimate")
```

Appendix 2 Reference:

Reference list

Bls.gov. (2024). *U.S. Bureau of Labor Statistics*. [online] Available at: https://www.bls.gov.

Borjas (2019). *Ise Labor Econ.* S.L.: Mcgraw-Hill Education.

Wooldridge, J. (2024). *Introductory Econometrics*.